10-01

Tourette's syndrome, a neurological disorder characterized by repetitive, involuntary movements, was named for Dr. Georges Gilles de la Tourette, the pioneering French neurologist who first described the condition in an eighty-six-year-old French noblewoman. It is almost always in childhood that the early

10-02

symptoms of Tourette's are first noticed, with the average onset between the ages of seven and ten years. It is estimated that two hundred thousand Americans have the most severe form of Tourette's, and as many as one in a hundred people exhibit milder and less complex symptoms that include chronic motor and

10-03

vocal tics or transient tics of childhood. For the majority of people diagnosed with the disorder, Tourette's is not a chronic condition with symptoms lasting a lifetime. Rather it is a condition whose severity peaks in the early teens, with improvement occurring in the late teens and continuing into adulthood. However, approximately

10-04

ten percent of those affected have a progressive or disabling course that lasts into adulthood. It is not uncommon for symptoms to be present for an extended period of time before a formal diagnosis of Tourette's is obtained. The reasons for this are many and may include the fact that for families and physicians

10-05

unfamiliar with Tourette's, mild and even moderate tic symptoms may be considered inconsequential, part of a developmental phase, or the result of another condition. For example, parents may think that eye blinking is related to vision problems or that sniffing is related to seasonal allergies. There are many

10-06

patients who are self-diagnosed after they, their parents, other relatives, or friends read or hear about Tourette's from others. Tics are classified as either simple or complex. Simple motor tics are sudden, brief, repetitive movements that involve a limited number of muscle

10-07

groups. Some of the more common simple tics include eye blinking, facial grimacing, shoulder shrugging, and head or shoulder jerking. Simple vocalizations, which motor tics typically precede, might include repetitive throat-clearing, sniffing, or grunting sounds. Complex tics are distinct, coordinated

10-08

patterns of movement that involve several muscle groups. Complex motor tics might include facial grimacing combined with a head twist and a shoulder shrug. Other complex motor tics may actually appear purposeful, including sniffing or touching objects, hopping, jumping, bending, or twisting.

10-09

Complex vocal tics include words or phrases. Perhaps the most dramatic and disabling tics include motor movements that result in self-harm such as punching oneself in the face or vocal tics including coprolalia, which involves uttering swear words, or echolalia, which involves repeating the words or phrases

10-10

of others. Tics are often worse with excitement or anxiety and better during calm, focused activities. Certain physical experiences can trigger or worsen tics. For example, tight collars may trigger neck tics, or hearing another person sniff or throat-clear may trigger the production of similar

09-13

However, such a scheme could not last unless someone was ultimately willing to pay such high prices and take possession of the bulbs. In February sixteen thirty-seven, tulip traders could no longer find new buyers willing to pay increasingly inflated prices for their bulbs. As this realization set

09-14

in, the demand for tulips collapsed, and prices plummeted. The speculative bubble burst. Some were left holding contracts to purchase tulips at prices now ten times greater than those on the open market, while others found themselves in possession of bulbs now worth a fraction of the price they had paid.

09-15

The event was popularized in eighteen forty-one by the book Extraordinary Popular Delusions and the Madness of Crowds, that a British journalist Charles Mackay wrote. That crowds of people often behave irrationally was the main thesis of the book, and tulip mania was one of the primary examples. According to

09-16

Mackay, at one point twelve acres of land were offered for a Semper Augustus bulb. He claims that many such investors ruined by the fall in prices were desperate, and Dutch commerce suffered a severe shock. In Mackay's account, the panicked tulip speculators sought help from the government of the Netherlands, which

09-17

responded by declaring that anyone who had bought contracts to purchase bulbs in the future could void their contract by payment of a ten percent fee. Attempts that were made to resolve the situation to the satisfaction of all parties were unsuccessful. The mania finally ended, Mackay says, with individuals stuck

09-18

with the bulbs they held at the end of the crash. No court would enforce payment of a contract, since judges regarded the debts as contracted through gambling, and thus not enforceable by law. Although Mackay's book is a classic that many publishing houses continue reprinting today, his account is contested. Many

09-19

modern scholars believe the mania was not as extraordinary as Mackay described, with some arguing that the price changes may not have constituted a bubble. Research on the tulip mania is difficult because of the limited data from the sixteen thirties. However, some modern economists have proposed rational explanations, rather than

09-20

a speculative mania, for the rise and fall in prices. For example, other flowers, such as the hyacinth, also had high prices on the flower's introduction, which then fell dramatically. The high prices may also have been driven by expectations of a parliamentary decree that contracts could be voided for a

09-21

small cost, thus lowering the risk to buyers. Despite these disagreements about whether the phenomenon technically constituted an economic bubble, the term "tulip mania" is still often used metaphorically to refer to any large economic bubble, when asset prices deviate from intrinsic

09-22

values.

09-03

United Provinces is generally thought to have started in earnest around fifteen ninety-three after the Flemish botanist Charles de l'~stecluse had taken up a post at the University of Leiden and established a botanical garden, which is famous as one of the oldest in the world. There, he planted his

09-04

collection of tulip bulbs, that the Emperor's ambassador sent to him from Turkey, which were able to tolerate the harsher conditions of the northern climate. It was shortly thereafter that the tulips began to grow in popularity. The flower rapidly became a coveted luxury item and a status symbol, and a profusion

09-05

of varieties followed. They were classified in groups: one-colored tulips of red, yellow, or white were known as Couleren, but it was the multicolored Rosen "ghred or pink on white background"ar, Violetten "ghpurple or lilac on white background"ar, and, to a lesser extent, the Bizarden

09-06

"ghred, brown or purple on yellow background"ar that were the most popular. These spectacular and highly sought-after tulip bulbs would grow flowers with vivid colors, lines, and flames on the petals, as a result, it is now understood, of being infected with tulip-specific virus known as the "Tulip

09-07

breaking virus". The Dutch Golden Age growers named their new varieties with exalted titles. Many early forms were prefixed with "admiral", often combined with the growers' names. "General" was another prefix that found its way into the names of around thirty varieties. Later came varieties with

09-08

even more superb names, such as even "Admiral of Admirals" and "General of Generals". Most of these varieties have now died out, though similar "broken" tulips continue in the trade. Tulips grow from bulbs, and can be propagated through both seeds and buds. Seeds from a tulip will form a flowering bulb

09-09

after seven-twelve years. When a bulb grows into the flower, the original bulb will disappear, but a clone bulb forms in its place, as do several buds. Properly cultivated, these buds will become bulbs of their own. The "tulip breaking" virus, which many botanists have studied, spreads only through buds, not seeds, and so

09-10

cultivating the most appealing varieties takes years. Propagation is greatly slowed down by the virus. Tulips bloom in April and May for about a week, and the secondary buds appear shortly thereafter. Bulbs can be uprooted and moved about from June to September, and thus actual purchases occurred during these

09-11

months. As the flowers grew in popularity, professional growers paid higher and higher prices for bulbs with the virus. By sixteen thirty-four, in part as a result of demand from the French, speculators began to enter the market. In sixteen thirty-six, the Dutch created

09-12

a type of formal futures markets which merchants used to sell and buy contracts to buy bulbs at the end of the season. Traders met at taverns and buyers were required to pay a fee per trade. People were purchasing bulbs at higher and higher prices, intending to re-sell them for a profit.